

MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

Artificial Intelligence and Society



Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 1.3: Algorithms for Supervised Learning

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A Simple Baseline

- **Frequency-based** guessing as the simplest algorithm for classification. Calculate probabilities of the classes P :

$$P(\text{good}) = 3/5$$

$$P(\text{bad}) = 2/5$$

→ classify **"good"**

- Improvement with **conditional probabilities** of the classes:

$$P(\text{red} \mid \text{good}) = 1/3$$

$$P(\text{young} \mid \text{good}) = 1/3$$

$$P(\text{red} \mid \text{bad}) = 1/2$$

$$P(\text{young} \mid \text{bad}) = 1/2$$

What quality is the unknown wine?

COLOR	AGE	QUALITY
red	old	good
white	old	good
white	young	good
red	old	bad
white	young	bad
red	young	?

$$\begin{aligned} &P(\text{good}) \cdot P(\text{red} \mid \text{good}) \cdot P(\text{young} \mid \text{good}) \\ &= 3/5 \cdot 1/3 \cdot 1/3 = \mathbf{1/15} \end{aligned}$$

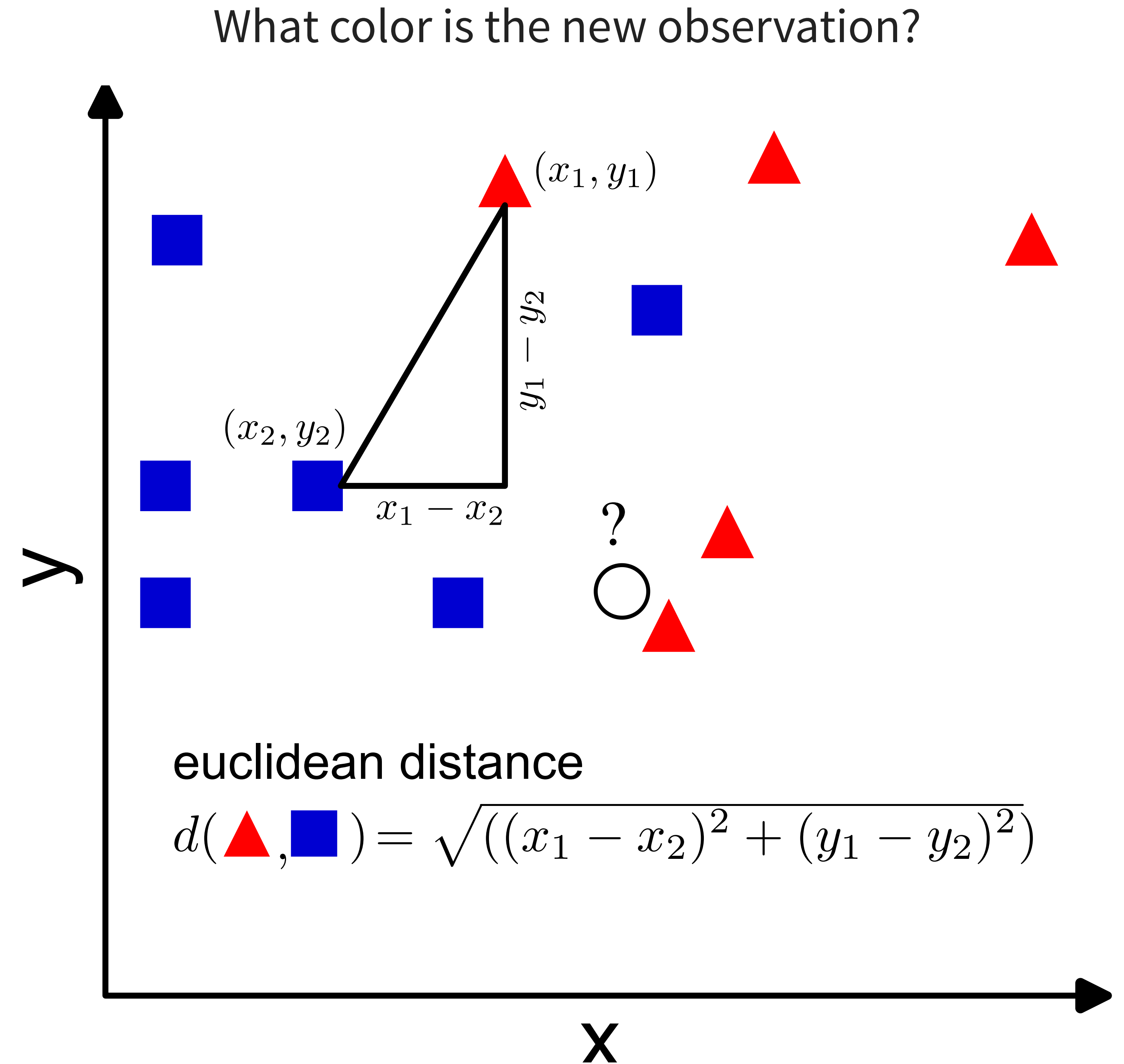
$$\begin{aligned} &P(\text{bad}) \cdot P(\text{red} \mid \text{bad}) \cdot P(\text{young} \mid \text{bad}) \\ &= 2/5 \cdot 1/2 \cdot 1/2 = \mathbf{1/10} \end{aligned}$$

$1/10$ is greater than $1/15$ → classify **"bad"**

Classification with conditional probabilities is based on Bayes' Theorem and is also called "Naive Bayes".

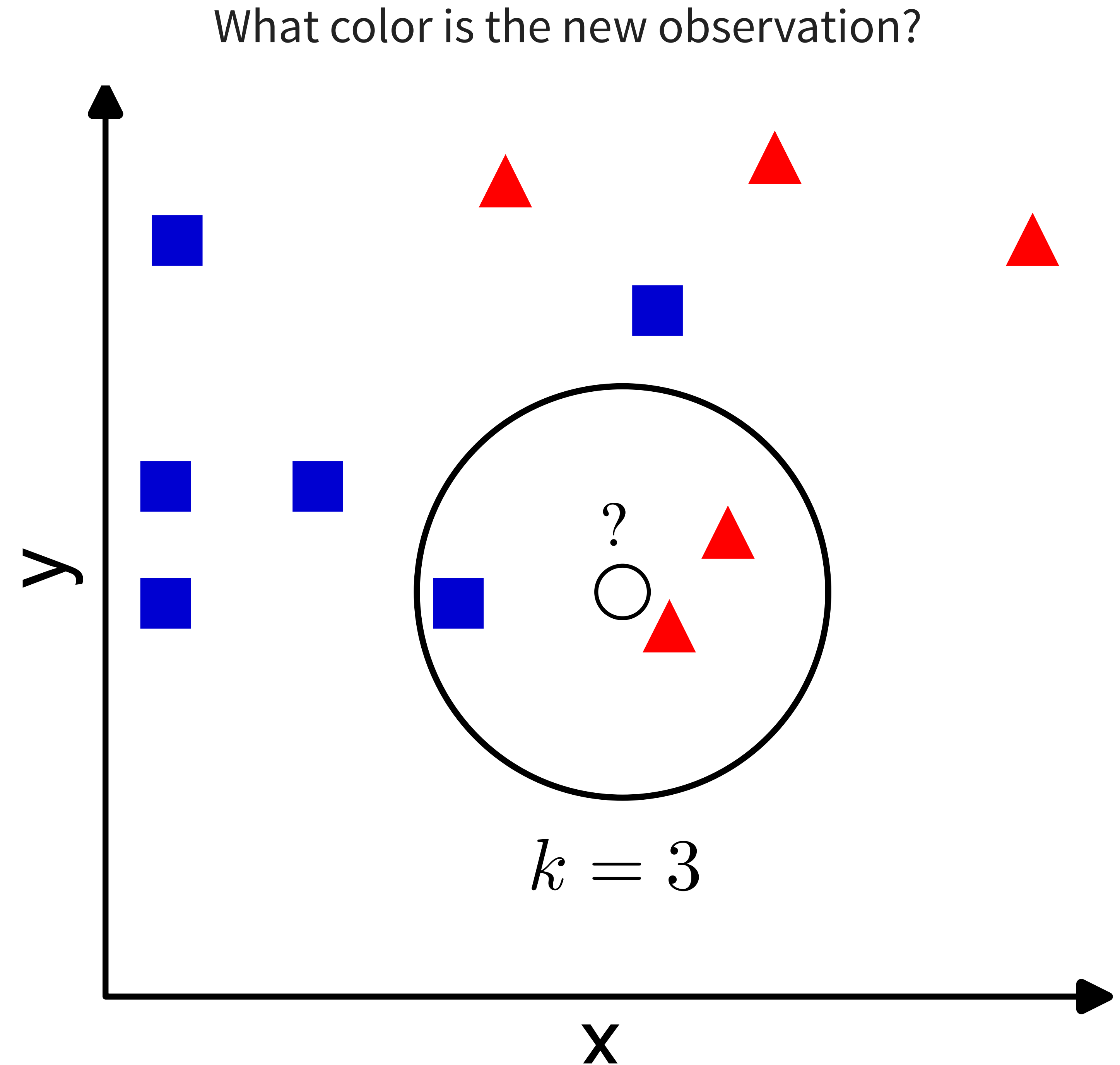
K Nearest Neighbors

- Define a **distance measure** between the data points (observations).



K Nearest Neighbors

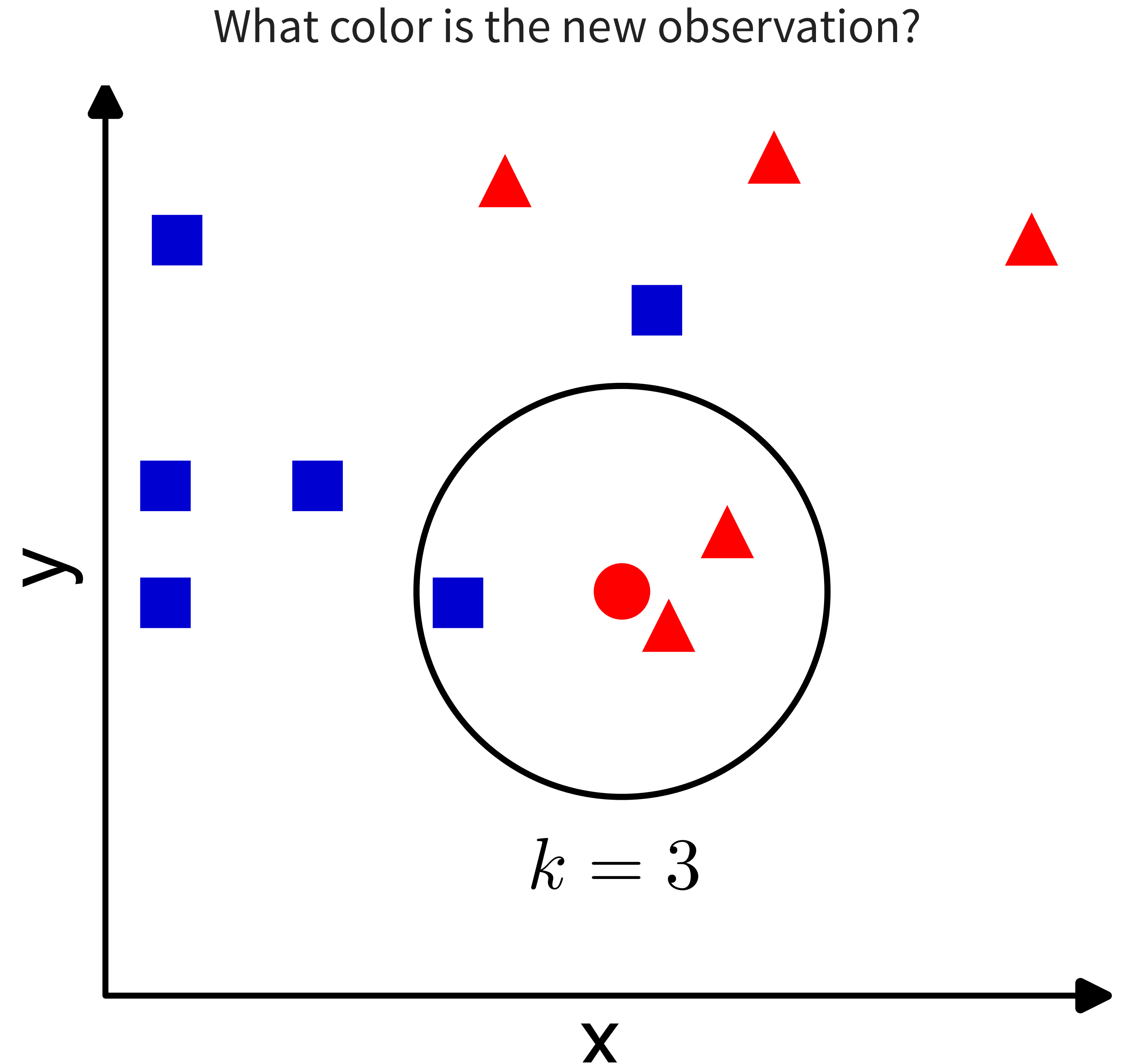
- Define a **distance measure** between the data points (observations).
- Define a number k .
- To classify a new observation:
 - Find the k nearest neighbors.



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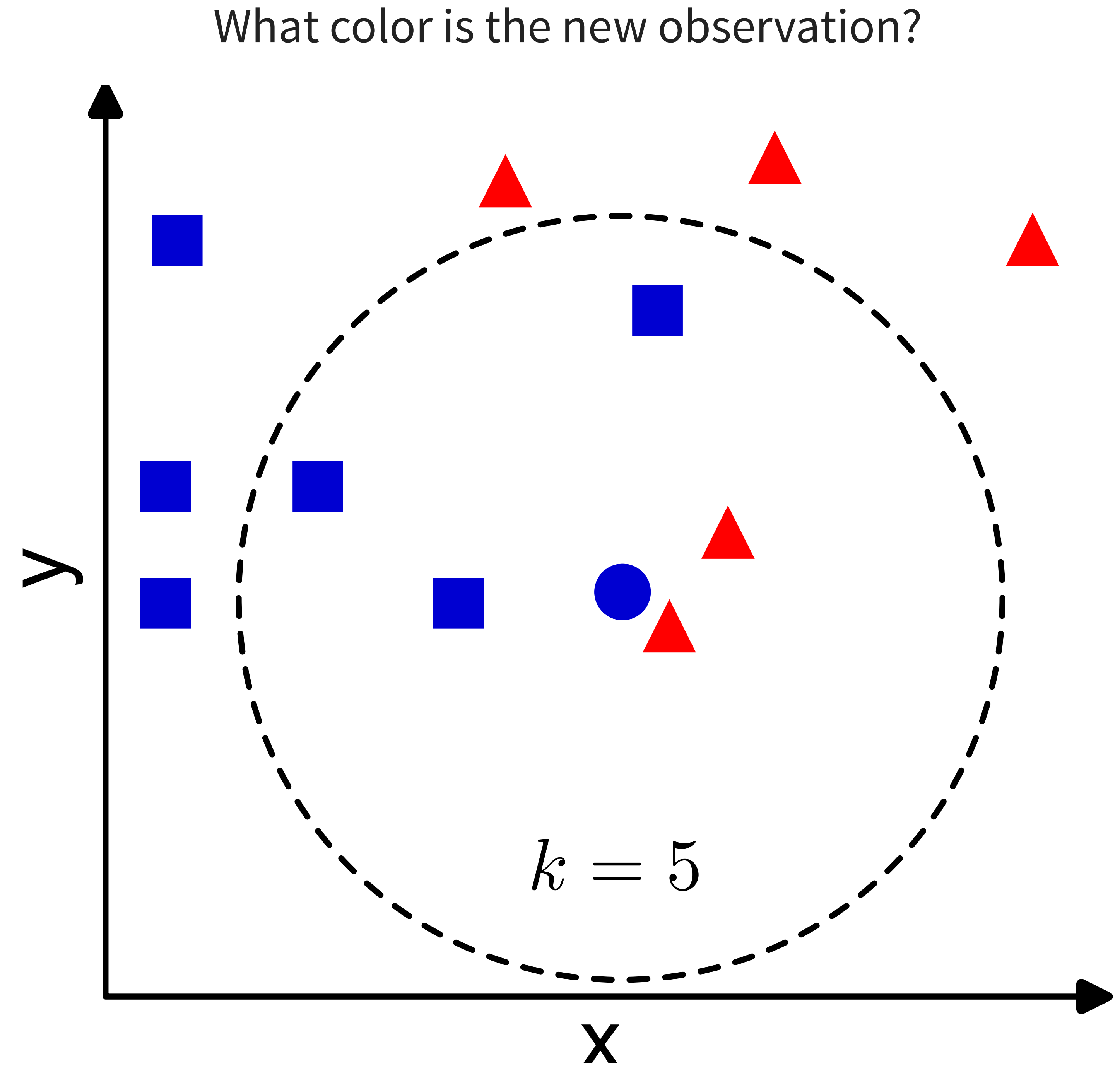
K Nearest Neighbors

- Define a **distance measure** between the data points (observations).
- Define a number k .
- To classify a new observation:
 - Find the k nearest neighbors.
 - Choose the class that the majority of the nearest neighbors has (majority vote).



K Nearest Neighbors

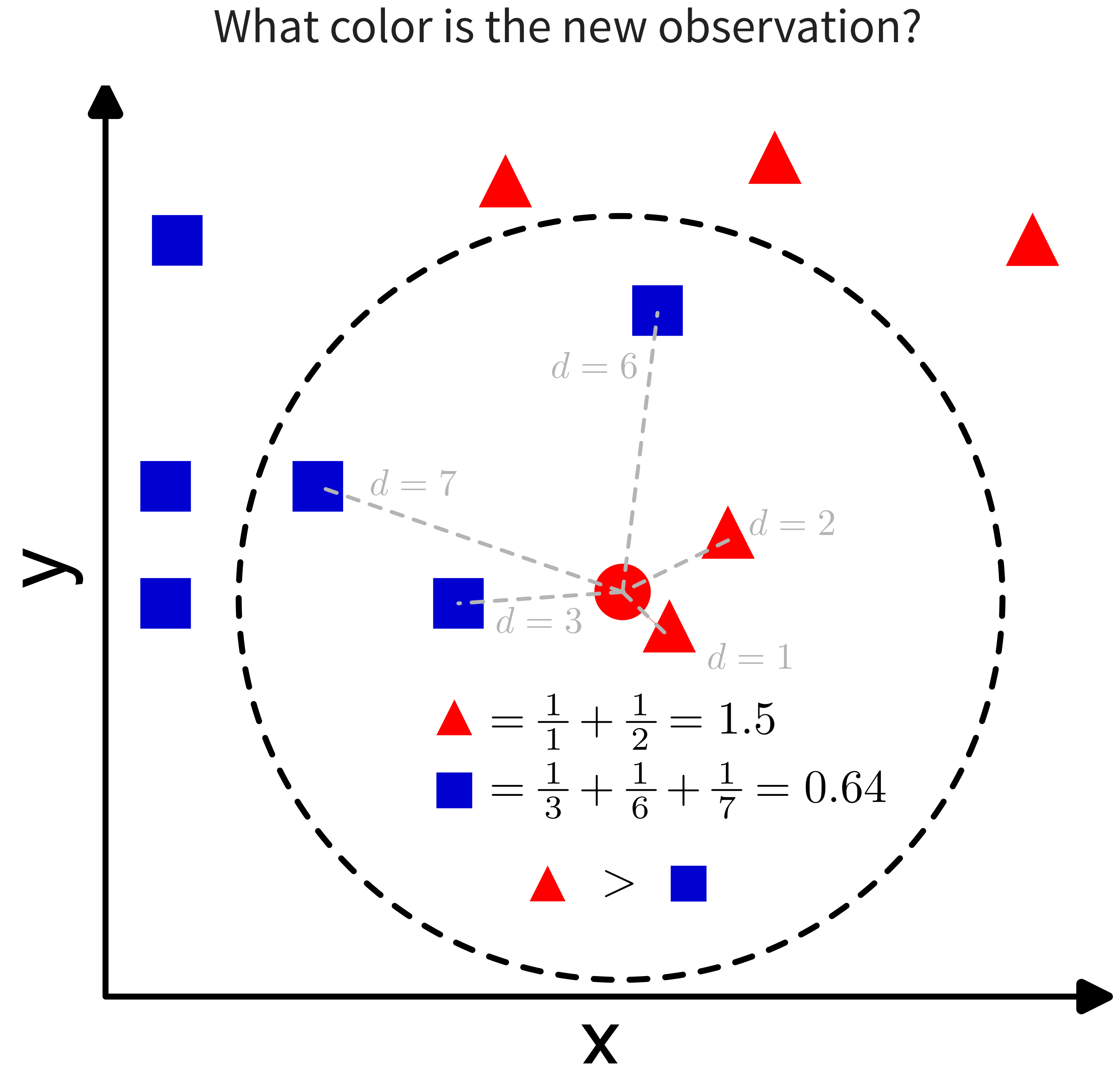
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K Nearest Neighbors

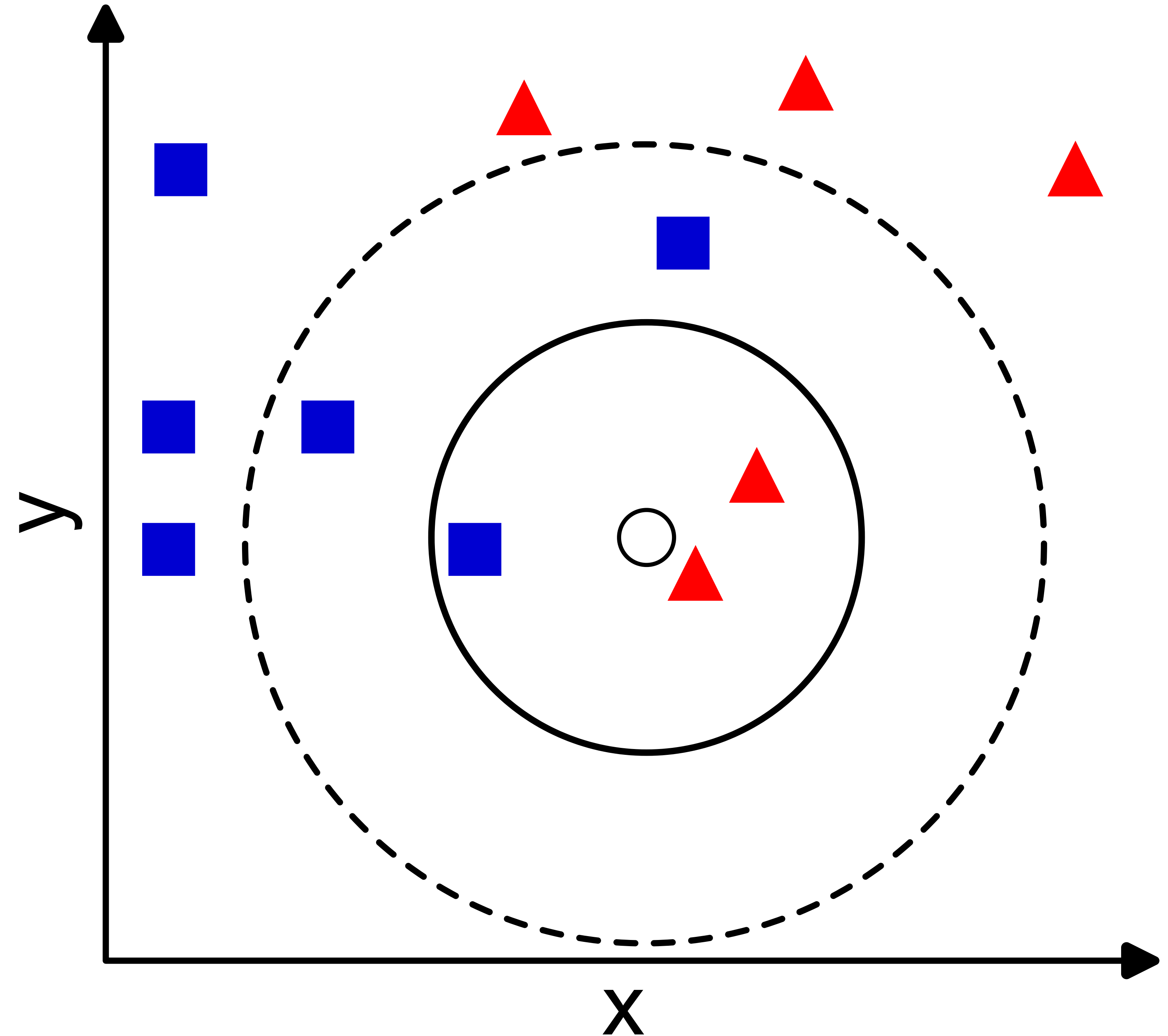
- Define a **distance measure** between the data points (observations).
- Define a number k .
- To classify a new observation:
 - Find the k nearest neighbors.
 - Choose the class that the majority of the nearest neighbors has (majority vote).
 - Optional: **weight** by the distance of the data points: closer neighbors have more influence.



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Some Observations about k Nearest Neighbors

- Depending on the definition of the algorithm (weighted vs. unweighted), the algorithm's result changes.
- Depending on the choice of k , the result also changes.
- k is also called a **hyperparameter** of the algorithm. Hyperparameters are used to control the algorithm's functionality.
- Hyperparameters must be adapted (tuned) to the characteristics of the dataset. In the example on the right, $k=12$ would be meaningless, as there are only 11 data points.

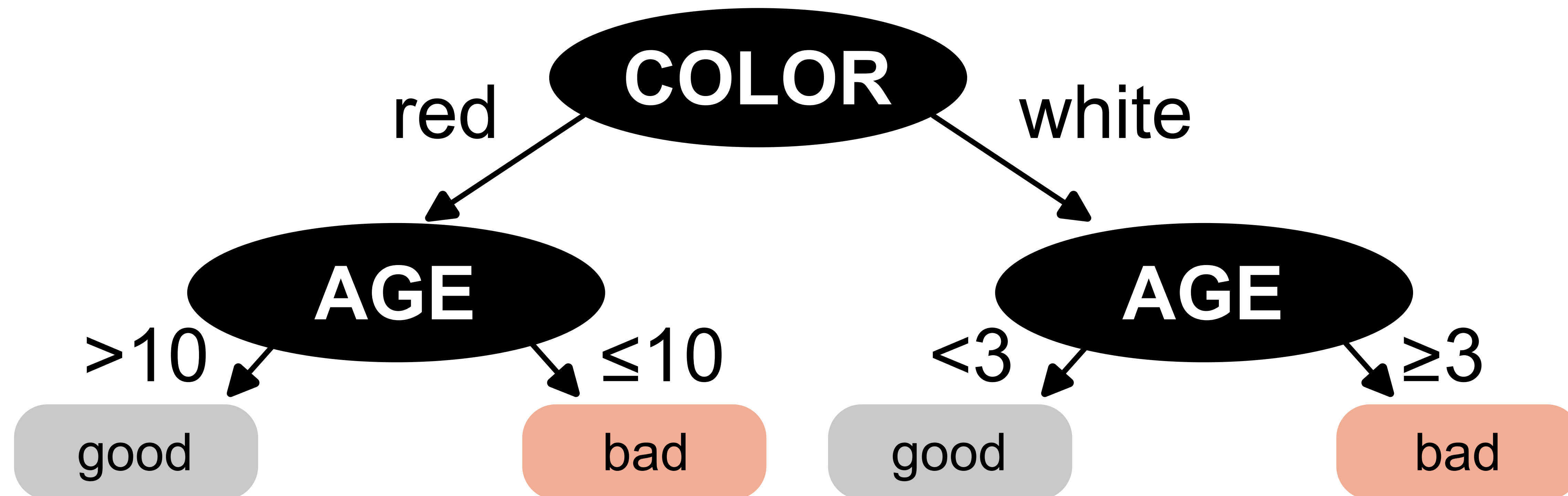


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Decision Trees

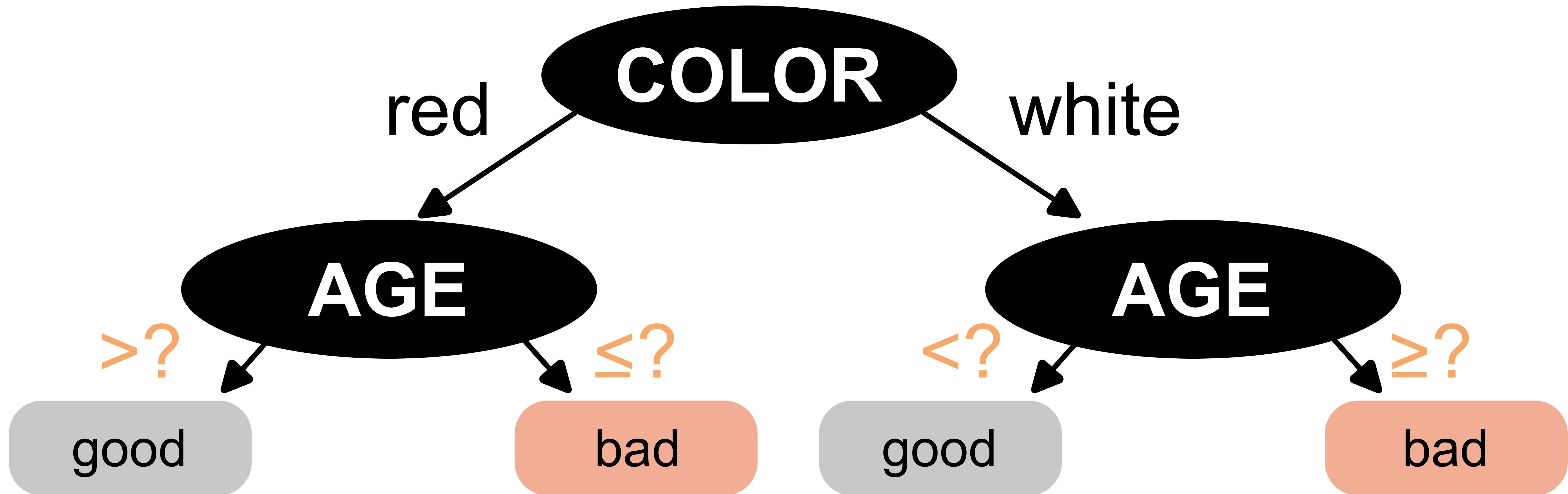
- Each **internal node** is an attribute.
- Each **edge** is a value (or range of values) of the source node.
- Each **leaf node** is a class label.
- Classification: Traversing the decision tree from top to bottom.

COLOR	AGE	QUALITY
red	12	good
white	13	good
white	2	good
red	10	bad
white	3	bad
red	3	?



Machine Learning

- So far, we have not processed any data to adapt the algorithm.
- We **train** the algorithm by adjusting the **thresholds** (parameters) based on the training data until the classes are optimally separated.



No Free Lunch Theorem

Which machine learning algorithm is best for classification?

Without having substantive information about the modeling problem, there is no single model that will always do better than any other model. Because of this, a strong case can be made to try a wide variety of techniques, then determine which model to focus on.

[David Wolpert, [The Lack of a priori Distinctions Between Learning Algorithms](#) (1996)]

Summary

- There are various algorithms for supervised learning. Examples include:
 - Naive Bayes
 - k nearest neighbours
 - Decision trees
- There is no "best" algorithm, but before a complicated algorithm is trained, it makes sense to have a simple **baseline** for comparison (e.g., Naive Bayes).
- "Learning" or **training** of an algorithm takes place by adjusting the parameters based on the training data until an optimal result is achieved.
- **Hyperparameters** (like the k in " k nearest neighbors") must be chosen such that they are meaningful for a given dataset.

Outlook

In the next set of inputs, we will learn about ...

- ... **neural nets** as a powerful architecture for machine learning algorithms. (Lecture 2.1)
- ... how to measure the **performance** of classification algorithms. (Lecture 2.2)
- ... **problems that can occur during algorithm training**, namely underfitting, overfitting, bias, and variance. (Lecture 2.3)
- ... more about **training data**, namely about different data types, and problems that can occur with data. (Lecture 2.4)
- ... an extension of supervised learning to **regression** – e.g. predicting numerical values instead of classes. (Lecture 2.5)